

Disorder as a Biophysical Prior for Active Learning in Protein Property Prediction

Bachelor Thesis Initial Presentation

Advisor: Sebastian Franz

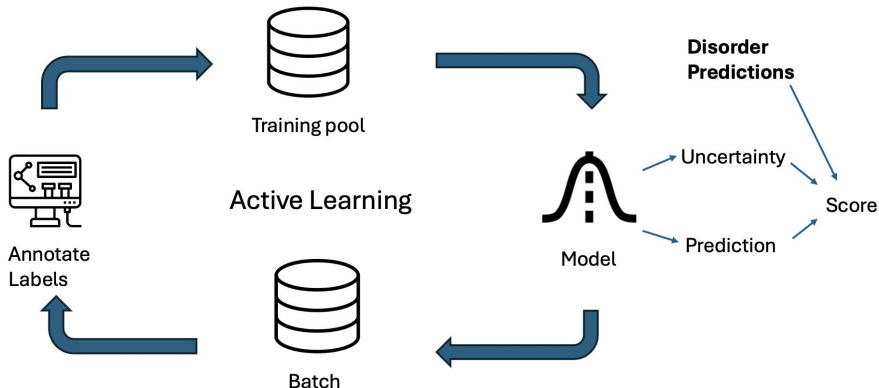
Supervisor: Prof. Burkhard Rost

Daryna Sova

April, 2026

Hypothesis

Adding a disorder term to the acquisition function of an active learning loop improves sample efficiency compared to uncertainty sampling alone.



Plan

- ▶ Validate disorder as a meaningful signal
- ▶ Test whether disorder contains predictive signal
- ▶ Design an experiment to incorporate disorder into active learning and test three different variants of acquisition function:
 1. uncertainty sampling
 2. pure disorder
 3. additive function (uncertainty+disorder)

Datasets

- ▶ Meltome (FLIP, curated): melting temperature [1]
- ▶ SCL: cellular localization [2]
- ▶ Amylase: DMS score/fitness [3]
- ▶ PHOT (single mutations): DMS score/fitness [4]

[1] J-SNACKKB. FLIP: Fitness Landscape Inference for Proteins. GitHub.

<https://github.com/J-SNACKKB/FLIP>

[2] Rostlab. Protein Benchmark Collection (PBC): Subcellular Localization (SCL). GitHub.

<https://github.com/Rostlab/pbc>

[3] Protein Engineering Tournament. PET Pilot 2023 In Vitro Dataset. GitHub.

<https://github.com/the-protein-engineering-tournament/pet-pilot-2023>

[4] Notin P.*, Kollasch A.W.*, Ritter D.*, van Niekerk L.*, Paul S., Spinner H., Rollins N., Shaw A., Weitzman R., Frazer J., Dias M., Franceschi D., Orenbuch R., Gal Y., Marks D.S.

ProteinGym: Large-Scale Benchmarks for Protein Fitness Prediction and Design.

Advances in Neural Information Processing Systems (NeurIPS), 2023.

https://proceedings.neurips.cc/paper_files/paper/2023/file/cac723e5ff29f65e3fcb0739ae91bee-Paper-Dataset_s_and_Benchmarks.pdf

Disorder Scores

- ▶ CheZOD – SETH [1]
- ▶ TriZOD – UdonPred [2]

[1] Ilzhöfer D, Heinzinger M and Rost B.

SETH predicts nuances of residue disorder from protein embeddings. Front. Bioinform. 2:1019597, 2022.
doi: 10.3389/fbinf.2022.1019597

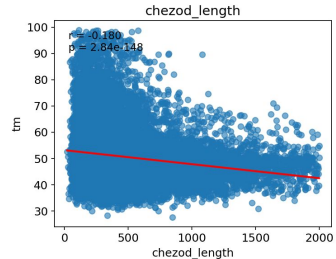
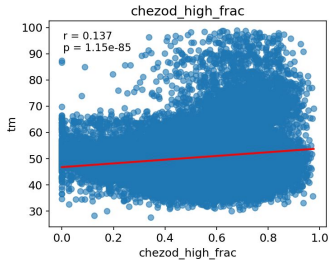
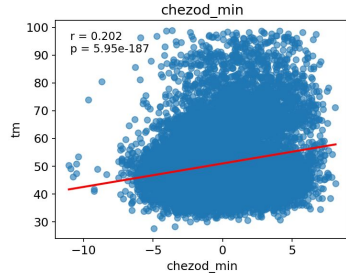
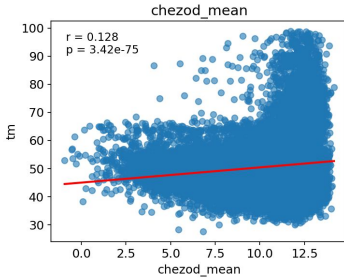
[2] Schlenzok, Julius and Wagemann, David and Senoner, Tobias and Haak, Markus and Rost, Burkhard.
UdonPred: Untangling Protein Intrinsic Disorder Prediction. bioRxiv, 2026.

doi: 10.64898/2026.01.26.701679

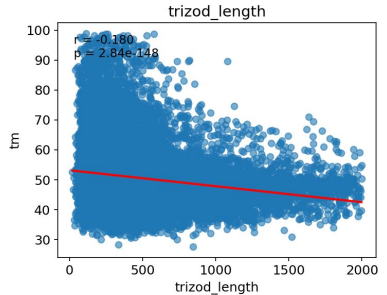
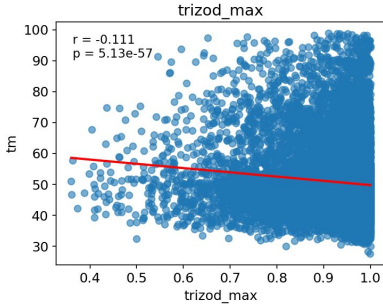
Validate Disorder Signal

- ▶ Compute per-residue disorder scores for the datasets and calculate per-sequence disorder values (mean, median, min, max, low fraction, high fraction) for correlation analysis
- ▶ Check correlation between target values and per-sequence disorder values

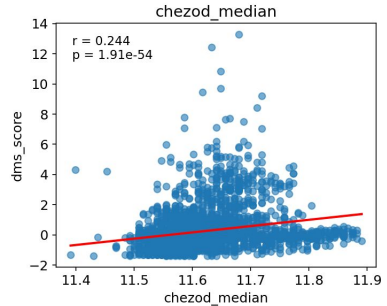
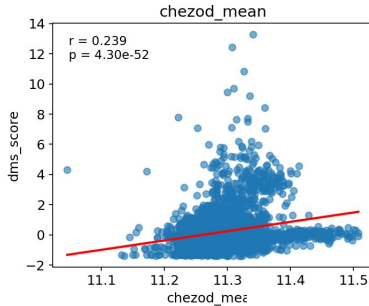
Meltome (CheZOD)



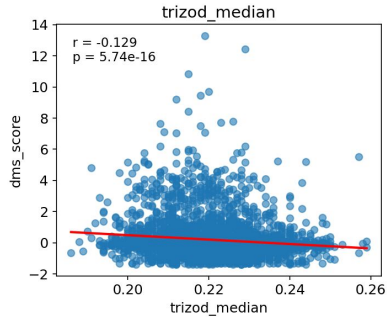
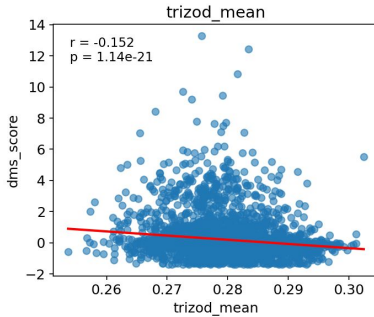
Meltome (TriZOD)



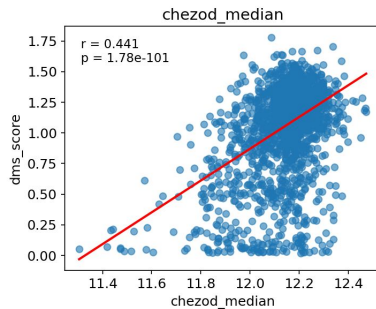
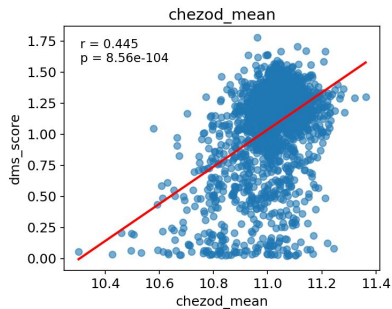
Amylase (CheZOD)

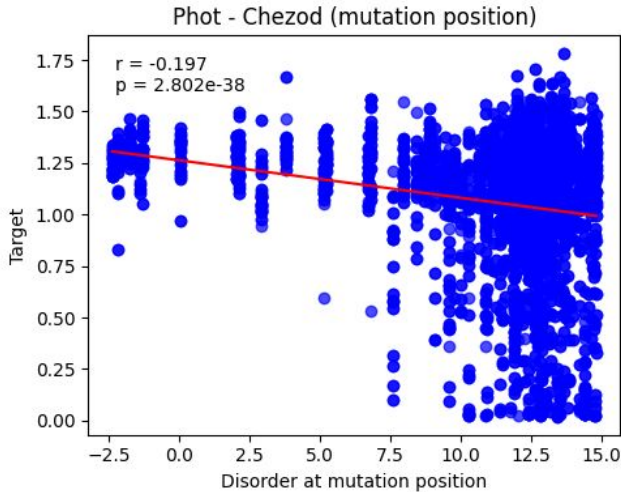


Amylase (TriZOD)

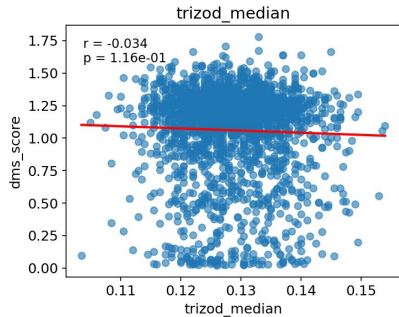
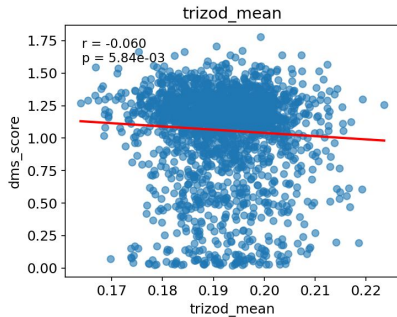


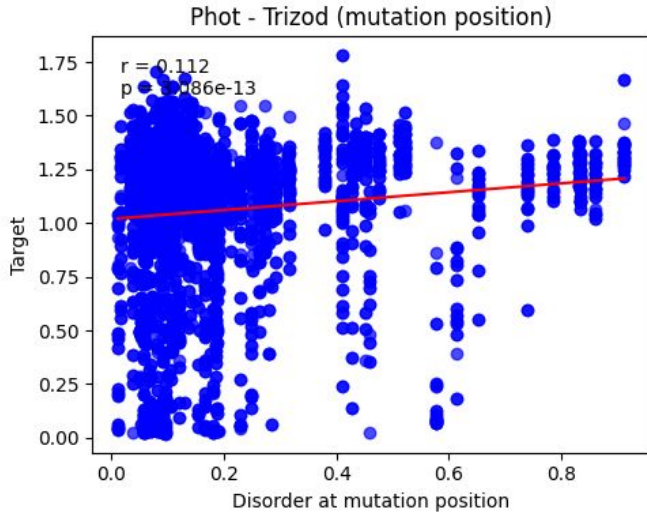
PHOT (CheZOD)



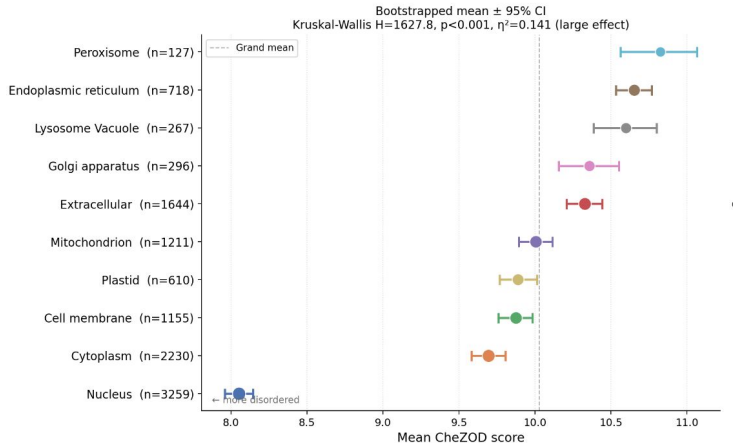


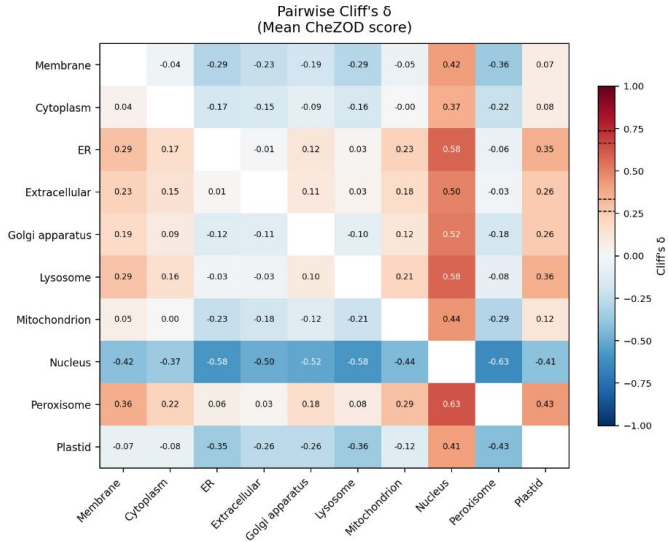
PHOT (TriZOD)



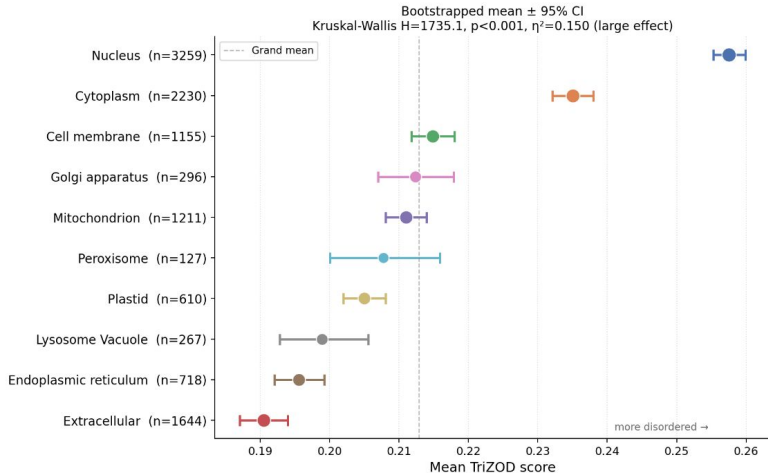


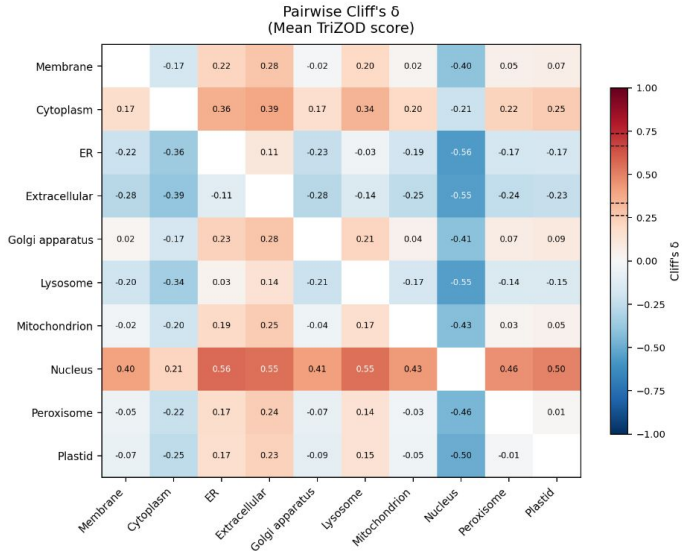
SCL Dataset Analysis (CheZOD)





SCL Dataset Analysis(TriZOD)





Summary

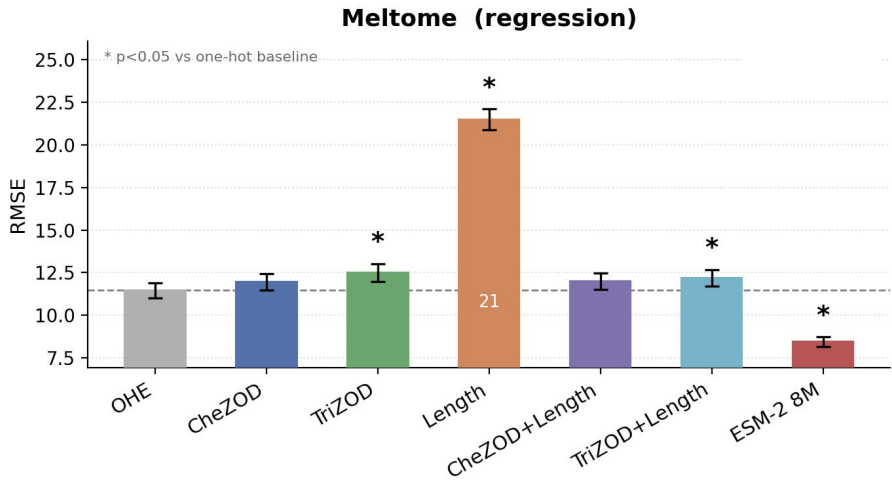
- ▶ Disorder shows correlation with protein features
- ▶ Disorder is a biologically meaningful signal for our tasks
- ▶ Disorder is a good candidate for a biophysical prior in AL

Does Disorder Contain Predictive Signal?

- ▶ Compare disorder-based inputs against one-hot baseline and pLM embeddings
- ▶ Evaluate with biotrainer [1]
- ▶ Reported metrics: RMSE, Accuracy, SCC, confidence interval, and significance versus one-hot

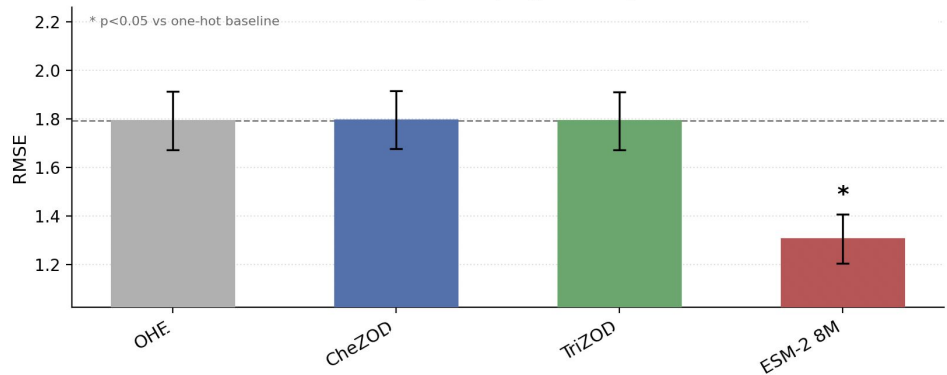
[1] Joaquin Gomez Sanchez and Sebastian Franz and Michael Heinzinger and Burkhard Rost and Christian Dallago. Standards, tooling and benchmarks to probe representation learning on proteins. NeurIPS 2022 Workshop on Learning Meaningful Representations of Life, 2022.
<https://openreview.net/forum?id=adODyN-eeJ8>

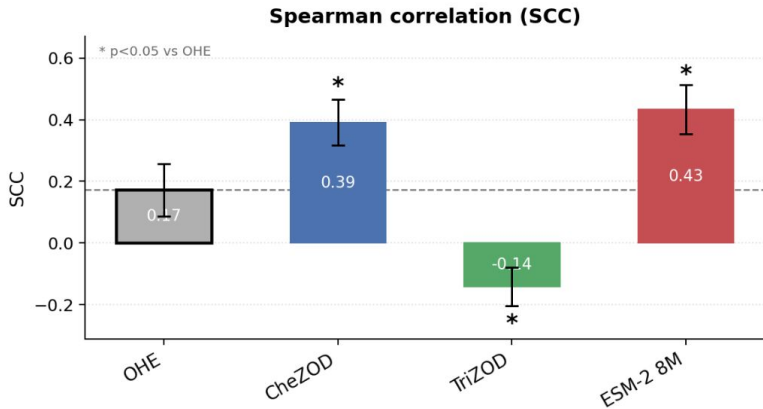
Meltome



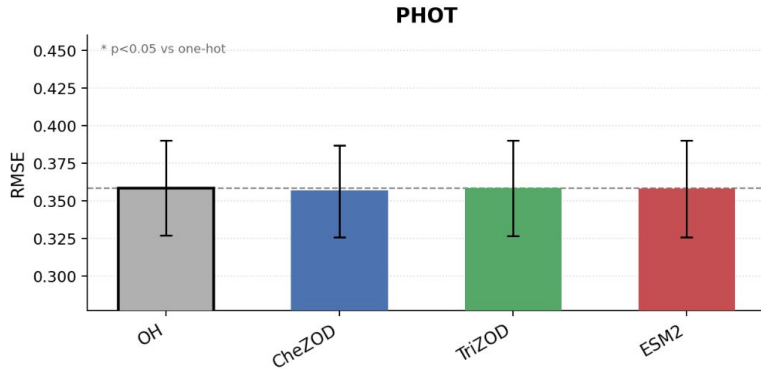
Amylase

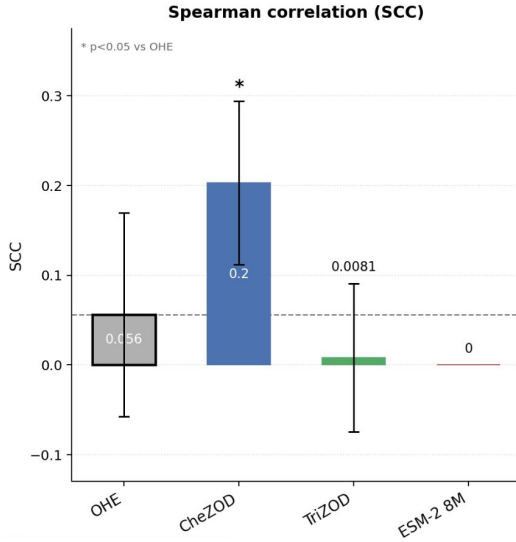
Amylase (regression)



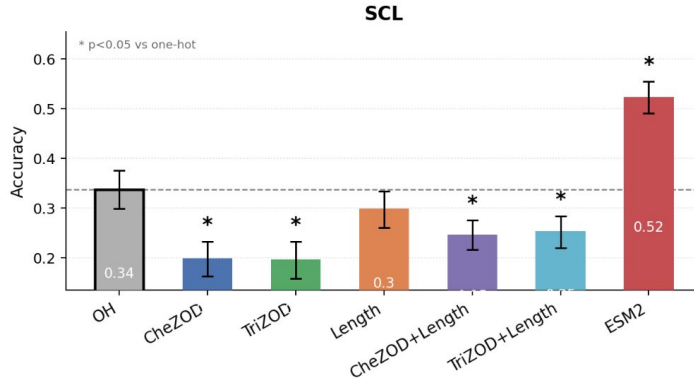


PHOT





SCL



Conclusion

- ▶ Disorder alone does not provide strong predictive signal. Weak predictive signal was observed only for DMS prediction tasks.
- ▶ Disorder shows limited predictive power, but remains biologically meaningful and can therefore be used as a biophysical informative prior in AL.
- ▶ Next step: design an experiment to incorporate disorder into active learning.

THANK YOU FOR YOUR ATTENTION

